AutoNAS: Automatic Neural Architecture Search for Hyperspectral Unmixing

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***Abstract*— Due to the powerful and automatic representation capabilities, deep learning (DL) techniques have made significant breakthroughs and progress in hyperspectral unmixing (HU). Among the DL approaches, autoencoders (AEs) have become a widely used and promising network architecture. However, these AE-based methods heavily rely on manual design and may not be a good fit for specific datasets. To unmix hyperspectral images more intelligently, we propose an automatic neural architecture search model for HU, AutoNAS for short, to determine the optimal network architecture by considering channel configurations and convolution kernels simultaneously. In AutoNAS, the self-supervised training mechanism based on hyperspectral images is first designed for generating the training samples of the supernet. Then, the affine parameter sharing strategy is adopted by applying different affine transformations on the supernet weights in the training phase, which enables finding the optimal channel configuration. Furthermore, on the basis of the obtained channel configuration, the evolutionary algorithm with additional computational constraints is introduced into networks to achieve flexible convolution kernel search by evaluating unmixing results of different architectures in the supernet. Extensive experiments conducted on four hyperspectral datasets demonstrate the effectiveness and superiority of the proposed AutoNAS in comparison with several state-of-the-art unmixing algorithms.**

***Index Terms*— Affine parameter sharing (APS), autoencoder (AE), deep learning (DL), evolutionary algorithm (EA), hyperspectral unmixing (HU), neural architecture search (NAS).**

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I. Introduction

BENEFITING from the rich spectral information and high spectral resolution, hyperspectral imagery (HSI) has received extensive attention in the field of Earth observation and remote sensing, such as dimensionality reduction [1]–[3], land cover classification [4]–[6], and target detection [7]–[9], which enables various ground objects to be effectively detected and identified [10]. However, due to the limited spatial resolution of the sensor, a single pixel is usually a mix of multiple materials in the scene, inevitably degrading the performance of high-level data processing [11]. Hyperspectral unmixing (HU) aims at solving this problem by separating mixed pixels into a combination of spectral signatures (endmembers) and their corresponding fractions (abundances) [12]–[14].

Unmixing methods currently depend on two types of mixing models to explain the interaction mechanism between the incident light and materials: the linear mixing model (LMM) and nonlinear mixing model (NLMM) [15]. LMM assumes a single interaction between the incident light and one material and has been widely used in the past decades due to its simplicity and flexibility. Nevertheless, LMM cannot consider the light's multiple scatterings among different materials in real-world remote sensing scenarios such as intimate mineral mixtures and vegetation canopies, which leads to observable unmixing errors [16], [17]. To address complex nonlinear interactions, numerous NLMMs have been proposed to model different order scattering effects and produce more accurate unmixing results [18]–[23], whereas the prior knowledge of order characteristics used to construct the NLMM is difficult to acquire in practice [24]. Once the adopted NLMM is not suitable for the actual scene, the accuracy of the estimated endmember and abundance results will be greatly reduced. Therefore, it is necessary to find a more generalized way to unmix HSI.

Inspired by the rapid development of deep learning (DL) techniques in pattern recognition and computer vision, an enormous number of learning-based approaches have been exploited in the area of HSI analysis, e.g., classification [25], feature learning [26], [27], and data fusion [28], [29]. The DL network is essentially a multilayer arithmetic structure stacked by many basic modules, which can learn complex nonlinear mapping relationships of the data. Due to its powerful learning and data fitting capabilities, DL has been widely used by a handful of works to address the unmixing problem.

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These DL networks mainly cover generative adversarial networks (GANs) [30], autoencoders (AEs) [31], convolutional neural networks (CNNs) [32], and recurrent neural networks (RNNs) [33]. Specifically, GAN can generate data samples with a specific distribution without knowing the probability distribution of the input data. According to this specialty, Tang *et al.* [34] introduced GAN to generate training samples with different dataset distributions for network training, which can exhibit competitive performance in unknown nonlinear scenes. The next architecture of interest for HU is the AE, which has already become the most explored network architecture, because it can automatically learn hidden representations and reconstruct data well. By imposing the nonnegativity, sum-to-one or sparsity constraints on the encoder part, the AE-based methods can provide good unmixing results superior to traditional methods [35]. Moreover, to incorporate spatial information from HSI, Zhang *et al.* [36] first proposed the deep CNN unmixing method, which proves the potential of CNNs in the task of unmixing. Hong *et al.* [37], for the first time, proposed a self-supervised spectral unmixing method in both pixelwise and patchwise manners, which effectively improves the unmixing performance without any ground-truth (GT) information. However, the CNN-based methods only consider local information of HSI and fail to explore the intrinsic properties from spectral dimensions, such as band-to-band variability and spectral correlation information. Although the aforementioned DL methods have achieved good unmixing performance, most of them rely on human experts to design network architectures. In this case, a lot of design time is consumed and the adopted architecture is not necessarily optimal.

Neural architecture search (NAS), as an essential task of automated machine learning, aims to design the network architecture with the best performance in an automated way. According to different search strategies, the current NAS methods can be roughly divided into three categories: gradient descent (GD), reinforcement learning (RL), and evolutionary algorithm (EA) [38]. With a growing demand for architecture performance, it is difficult to meet the requirements by relying on manually designed architectures. Especially for more complex remote sensing datasets, the object features extracted by different network architectures usually vary significantly. If some high-level features are ignored in the handcrafted architecture, it will have a great impact on the final recognition accuracy [39]. The emergence of NAS solves this pain point, and a large number of NAS studies have been applied in the field of remote sensing. Jing *et al.* [40] put forward a GD-based NAS method for remote sensing image scene classification and adopted edge normalization to make the architecture search more efficient. To control the complexity of the network architecture, a multiobjective scene classification NAS method based on EA was proposed to balance the network performance and computational efficiency [41]. In [42], an automatic CNN method based on GD was designed for LiDAR data classification and the improved label smoothing with student's t-distribution was introduced to further mitigate the overfitting problem. Michel *et al.* [43] proposed a novel policy gradient search strategy for hyperspectral band selection, yielding satisfying results. Zhang *et al.* [44] employed a cell-based CNN architecture search method by particle swarm optimization in the hyperspectral image classification task, which is capable of obtaining the global optimal architecture compared with the GD-based method. Nevertheless, these existing NAS methods tend to search for network connections and convolution kernels rather than the number of channels. In fact, the search for the number of channels is also important since the channel configuration directly affects the affordability of the network on resource-constrained platforms [45]. For example, in a single fully connected or convolution layer, the floating-point operation per second (FLOPS) increases linearly by the output channel. Therefore, the search for the optimal channel configuration should be considered on the basis of existing NAS methods.

To overcome the aforementioned challenge, we aim to build a complete automatic NAS framework for HU, called AutoNAS. For this purpose, we first generate self-supervised training sets based on HSI for the network search. Then, an affine parameter sharing (APS) method is proposed to obtain the optimal channel configuration by applying different affine transformations on network weights. On the basis of the obtained channel configuration, the EA-based method is adopted to search for the network convolution kernel, which implements a more flexible extraction of HSI. More specifically, the main contributions of this article are highlighted in the following.

1) We propose an automatic NAS framework for HU tasks, called AutoNAS, to determine the optimal network architecture by considering channel configuration and convolution kernels simultaneously. AutoNAS is capable of achieving more automated and intelligent HU and meanwhile greatly reducing labor costs while maintaining optimal unmixing performance.
2) Self-supervised training sets are prepared and constructed to assist the subsequent NAS method. Currently, the training of the NAS method is basically dominated by the supervised approach, but it is hard to obtain the GT information in HU due to the difficulty of real measurements. By extracting the endmembers and computing their abundances in HSI by the means of model-driven methods, we can avoid the GT acquisition and enable the search network to learn effective mapping relations.
3) The APS method is devised to search the number of channels in the classic AE-based unmixing framework. NAS methods usually adopt parameter sharing techniques to reduce redundant training, yet the high-level parameter sharing might reduce the network's discrimination between different architectures. APS is capable of dynamically adjusting the tradeoff between efficient training and architecture discrimination so that the entire search process achieves optimal performance. To the best of our knowledge, this is the first time to explore the automatic design of the channel search for the HU task.
4) The EA-based method with additional computational constraints is customized for obtaining the flexible and accurate solution of convolution kernels. Also, the

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network accuracy and computational complexity are optimized at the same time.

The remainder of this article is organized as follows. The related research work is presented in Section II. The proposed AutoNAS framework is introduced in Section III. Section IV describes extensive experiments on four hyperspectral datasets in this study. A discussion and our conclusions are summarized in Section V.

II. Related Research Work

A. AE-Based Unmixing Networks

The AE unmixing network can be regarded as a typical data-driven method in an unsupervised fashion. It can achieve desirable performance by automatically learning hidden representations (e.g., the abundance) and reconstructing the data with the corresponding base (e.g., the endmember) [46]. In general, the objective function of the AE-based unmixing method is to minimize the reconstruction error in different measurement forms, such as mean square error (mse), spectral angle distance (SAD), and spectral information divergence (SID). Meanwhile, the abundance's nonnegative and sum-to-one constraints are imposed on the hidden representation layer to facilitate the training of the AE unmixing network [47]. In addition, to meet various functional requirements, a series of improvements has been applied to the AE unmixing framework. For instance, stacked nonnegative sparse and denoising AE unmixing methods have been proposed to address the issue of outliers and noisy data [48]–[50]. In [51]–[53], CNNs have been introduced into the traditional AE unmixing architecture to further enhance the unmixing accuracy by integrating the spatial structure of HSI. With the rapid development of different imaging techniques, LiDAR can be adopted as the external multimodal data to improve unmixing performance [54]. Moreover, in order to enrich the training sample characteristics of the AE unmixing network, a number of self-supervised technologies have been designed to assist network learning and reduce information loss [55], [56]. As for complex nonlinear scenes, Wang *et al.* [57] adopted a well-designed decoder architecture to estimate the second-order scattering interactions, which proves the effectiveness of the AE-based method for nonlinear HU. Furthermore, multitask learning technology was developed to improve the unmixing performance by disassembling the nonlinear unmixing process into different tasks and utilizing shared information among them [58].

Although the above typical AEs have made a lot of breakthroughs in HU, they mainly acquire fixed image information under the fixed network architecture and connection modes and are not driven by the characteristics of the input image. Thus, in this article, an AutoNAS framework for AE-based unmixing networks is investigated.

B. Principles of NAS

NAS consists of three main components: search space, search strategy, and performance estimation. Fig. 1 shows the general framework of NAS for HSI. NAS usually starts with a set of candidate operations and adopts a search strategy to obtain substantial candidate architectures based on the search space. Then, these candidate architectures are trained from scratch on the training set and evaluated according to their performance on the validation set. The evaluation information of candidate architectures is considered feedback information to regulate the search strategy to obtain new candidate architectures. Eventually, when the search is terminated, the top-ranked architecture is selected as the optimal network architecture.

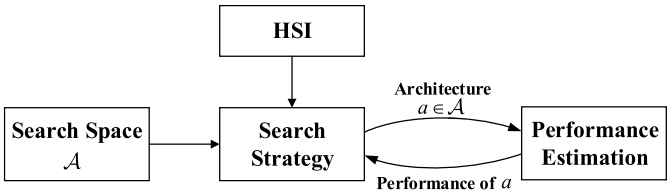


Fig. 1. General framework of NAS for HSI.

Early NAS work roughly followed the above process. However, training each candidate architecture from scratch consumes a lot of computing resources. To address this weakness of NAS, parameter sharing is widely applied to improve the search efficiency by utilizing the trained parameters [59]. One strategy is to train a supernet containing all candidate operations in the search space, and the shared weights in the supernet can accelerate the training of these candidate architectures [60]–[62]. Another strategy is to treat different candidate architectures as trainable variables and guide them to be fully trained by transferring the weights of existing successful neural networks [63]–[65]. Although these two sharing strategies can perform well in some scenarios, the mechanism of parameter sharing in the search process is still rarely explored. Recently, Wang *et al.* [66] first analyzed the positive and negative effects of different parameter sharing levels for the network search. It was concluded that the high-level sharing speeds up the search process while reducing the ability to discriminate between different candidate architectures. To balance search efficiency and architecture discrimination, a general parameter sharing scheme based on affine transformation was further proposed to realize effective search. In the field of HSI, existing NAS methods do not consider the channel configuration in the search space and instead apply human-crafted heuristics [67], [68]. Therefore, inspired by the affine transformation idea, we present the APS strategy to search for the number of channels in the proposed AutoNAS framework and the specific introduction is illustrated in Section III-B.

III. AutoNAS Framework for HU

In this section, we will detail the unmixing process of the proposed AutoNAS framework step by step. Fig. 2 presents the overall flowchart of the designed AutoNAS framework, including four procedures to search for the optimal AE-based unmixing architecture: construction of data and search spaces, channel search method based on APS, kernel search method based on EA, and unmixing performance evaluation.

A. Construction of Data and Search Spaces

Due to the lack of GT in HU, a series of training and validation data is designed and generated through the method

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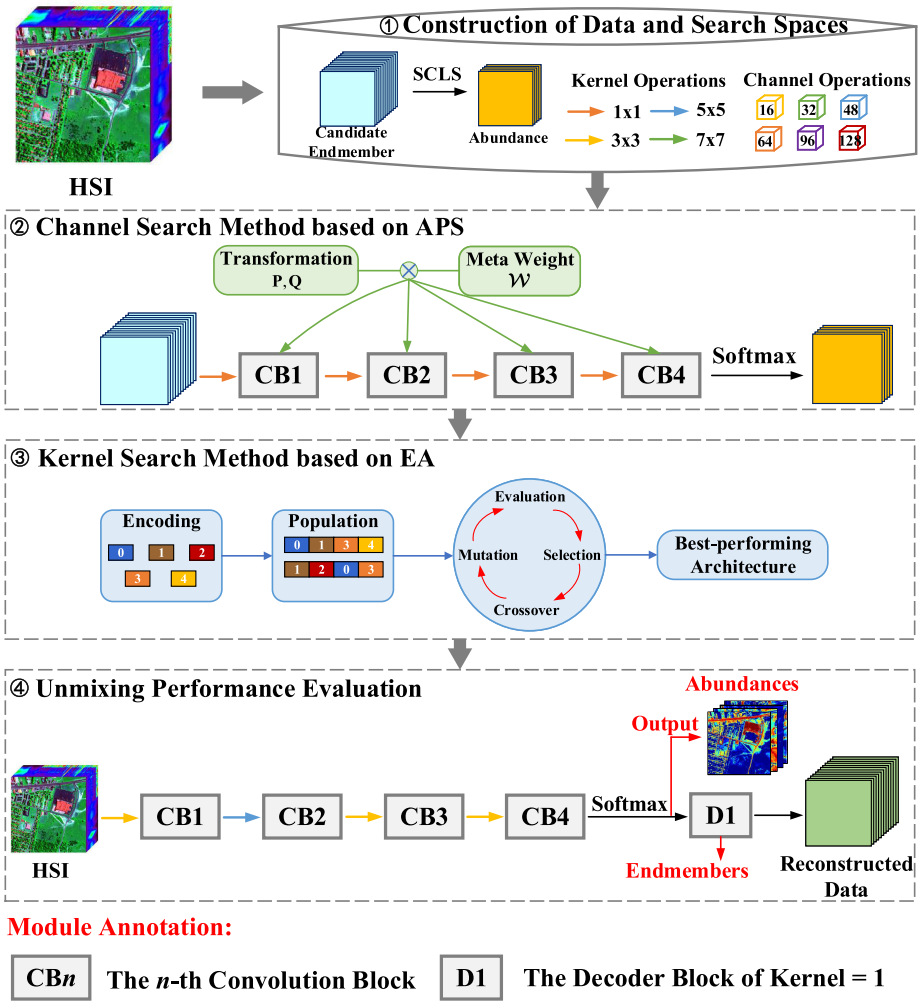


Fig. 2. Overall flowchart of the proposed AutoNAS framework.

of our previous work [55]. Specifically, the original HSI is first divided into some subsets and the vertex component analysis (VCA) [69] is applied in each subset to extract a certain number of candidate endmembers, denoted as $\mathbf{S}_1 \in \mathbb{R}^{L\times K}$ with L bands and K candidate endmembers. The corresponding abundances $\mathbf{A}_1 \in \mathbb{R}^{S\times K}$ with S endmember categories are obtained by the classic LMM-based method, such as the scaled constrained least squares (SCLS) method [70]. By learning the mapping relation from candidate endmembers to their abundances, the supernet has been fully trained to help the subsequent acquisition of the optimal architecture on the validation set. In essence, this self-supervised training strategy can achieve a similar effect to the strategy that considers the original HSI and true abundances. The advantage is that it can avoid additional information collection and effectively solve the problem of few training samples in HU.

The search space in the proposed AutoNAS contains two parts: kernel operations and channel operations. Since most existing AE unmixing methods are single-path architecture, we do not consider the search for network connections in this article. The kernel operations consist of four possible operations: 1×1 convolution, 3×3 convolution, 5×5 convolution, and 7×7 convolution, and the channel operations $\mathcal{C}$ are set to $\{16, 32, 48, 64, 96, 128\}$. Note that the block-based encoding space is adopted in the search process, called convolution block, where different layers of various types are assembled as the basic unit of the search space. The convolution block includes rectified linear unit (ReLU), convolution, and batch normalization (BN) in turn. The advantage of the block-based encoding space is that the number of parameters is greatly reduced, and as a result, the search network can have promising performance.

B. Channel Search Method Based on APS

Channel search methods aim to find the optimal number of channels in each layer of a network. Here, the parameter sharing scheme is adopted to reduce the computational cost by encoding the entire search space into a single supernet, and all candidate architectures are shared by network weights $\mathcal{W}$ of the supernet during training. Then, the optimal channel is determined based on the performance ranking on the validation set. The channel search problem can be formulated as

$$\begin{aligned} \mathcal{W}^* &= \arg\min_{\mathcal{W}} \mathcal{L}_{\text{train}}(f(c_1, c_2, \ldots, c_l; \mathcal{W})) \\ &\text{s.t. } \mathcal{B}(\mathcal{W}) \le B \end{aligned} \tag{1}$$

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$$(c_1, c_2, \ldots, c_l)^* = \underset{c_1,c_2,\ldots,c_l}{\arg\min}\, \mathcal{L}_{\text{val}}(f(c_1, c_2, \ldots, c_l; \mathcal{W}^*)) \quad (2)$$

where $\mathcal{L}_{\text{train}}$ and $\mathcal{L}_{\text{val}}$ denote the training and validation loss, respectively, such as mse between the estimated abundances $\hat{\mathbf{A}}_1$ and true abundances $\mathbf{A}_1$ for candidate endmembers. f represents the channel search network and $(c_1, c_2, \ldots, c_l)$ denotes the channel decision with l layers. $\mathcal{B}(\mathcal{W})$ is the network cost function (i.e., FLOPS) and B is the computational constraint.

However, the optimal network weight $\mathcal{W}^*$ does not necessarily guarantee that different candidate architectures can be effectively discriminated on the validate set because these candidate architectures may be not fairly and fully trained [65]. To achieve a better balance between efficient training and architecture discrimination, APS is adopted into the channel search method. Specifically, suppose that c_{in} and c_{out} are the input and output channel of the lth convolutional layer, and the network weight $\mathbf{W}^{c_{\text{out}},c_{\text{in}}} \in \mathbb{R}^{c_{\text{out}}\times c_{\text{in}}\times m\times m}$ is expressed by affine transformation as

$$\mathbf{W}^{c_{\text{out}},c_{\text{in}}} = \left(\mathbf{Q}^{c_{\text{out}}}\right)^T \times_2 \mathcal{W} \times_1 \mathbf{P}^{c_{\text{in}}} \quad (3)$$

where $\times_n$ denotes the n-mode matrix product [71]. $\mathbf{P}^{c_{\text{in}}} = [\mathbf{p}_1, \ldots, \mathbf{p}_{c_{\text{in}}}] \in \mathbb{R}^{c\times c_{\text{in}}}$ and $\mathbf{Q}^{c_{\text{out}}} = [\mathbf{q}_1, \ldots, \mathbf{q}_{c_{\text{out}}}] \in \mathbb{R}^{c\times c_{\text{out}}}$ $(c \geq c_{\text{in}}, c_{\text{out}})$ are both standard semiorthogonal transformation matrices. $\mathcal{W} \in \mathbb{R}^{c\times c\times m\times m}$ is the meta-weight as the shared parameter pool and follows the standard normal distribution, where c and m represent the number of filters and kernel size, respectively.

The level of parameter sharing is calculated as the Frobenius norm of cross-covariance matrix between different network weights $\mathbf{W}^{j,i}$ and $\mathbf{W}^{\hat{j},\hat{i}}$, which is given by

$$\begin{aligned}\mathcal{L}_p &= \left\|\text{Cov}\left(\mathbf{W}^{j,i}, \mathbf{W}^{\hat{j},\hat{i}}\right)\right\|_F^2 \\ &= \sum_{x=1}^{c_i}\sum_{y=1}^{c_j}\sum_{\hat{x}=1}^{c_{\hat{i}}}\sum_{\hat{y}=1}^{c_{\hat{j}}}\left[E\left(\mathbf{W}_{y,x}^{j,i}\mathbf{W}_{\hat{y},\hat{x}}^{\hat{j},\hat{i}}\right) - E\left(\mathbf{W}_{y,x}^{j,i}\right)E\left(\mathbf{W}_{\hat{y},\hat{x}}^{\hat{j},\hat{i}}\right)\right]^2\end{aligned} \quad (4)$$

where Conv(.) and $E(.)$ denote the function values of the cross covariance and expectation for the network parameters, respectively. Since the elements in $\mathbf{P}$ and $\mathbf{Q}$ are both standard orthogonal basis, the second term in (4) can be calculated as

$$E\left(\mathbf{W}_{y,x}^{j,i}\right) = E\left(\left(\mathbf{q}_y\right)^T\mathcal{W}\mathbf{p}_x\right) = \left(\mathbf{q}_y\right)^T E(\mathcal{W})\mathbf{p}_x = 0 \quad (5)$$

$$E\left(\mathbf{W}_{\hat{y},\hat{x}}^{j,i}\right) = E\left(\left(\mathbf{q}_{\hat{y}}\right)^T\mathcal{W}\mathbf{p}_{\hat{x}}\right) = \left(\mathbf{q}_{\hat{y}}\right)^T E(\mathcal{W})\mathbf{p}_{\hat{x}} = 0. \quad (6)$$

We further rewrite (4) as follows:

$$\begin{aligned}\mathcal{L}_p &= \sum_{x=1}^{c_i}\sum_{y=1}^{c_j}\sum_{\hat{x}=1}^{c_{\hat{i}}}\sum_{\hat{y}=1}^{c_{\hat{j}}}\left[E\left(\mathbf{W}_{y,x}^{j,i}\mathbf{W}_{\hat{y},\hat{x}}^{\hat{j},\hat{i}}\right)\right]^2 \\ &= \sum_{x=1}^{c_i}\sum_{y=1}^{c_j}\sum_{\hat{x}=1}^{c_{\hat{i}}}\sum_{\hat{y}=1}^{c_{\hat{j}}}\left[\left(\mathbf{q}_y\right)^T E\left[\mathcal{W}\mathbf{p}_x(\mathbf{p}_{\hat{x}})^T\mathcal{W}^T\right]\mathbf{q}_{\hat{y}}\right]^2 \\ &= \sum_{x=1}^{c_i}\sum_{y=1}^{c_j}\sum_{\hat{x}=1}^{c_{\hat{i}}}\sum_{\hat{y}=1}^{c_{\hat{j}}}\left[\left(\mathbf{q}_y\right)^T\mathbf{q}_{\hat{y}}\cdot(\mathbf{p}_x)^T\mathbf{p}_{\hat{x}}\right]^2 \\ &= \left\|\left(\mathbf{Q}^j\right)^T\mathbf{Q}^{\hat{j}}\right\|_F^2 \cdot \left\|\left(\mathbf{P}^i\right)^T\mathbf{P}^{\hat{i}}\right\|_F^2.\end{aligned} \quad (7)$$

With the increase of $\mathcal{L}_p$, the channel search network can achieve more efficient searching, but a larger coupling among different candidates is generated, leading to low discriminant performance during the validation phase. Therefore, the objective function of APS is designed to contain two stages for the channel search. In the first stage, we maximize $\mathcal{L}_p$ to speed up the searching process by initializing $\mathbf{P}^i$ and $\mathbf{Q}^j$ with the identity matrix, and the LSTM controller $\pi(\theta)$ uniformly samples the channel decision $a_c = [c_1, \ldots, c_l] \sim \pi(\theta)$ under the fixed parameters θ. In this process, $\mathcal{W}$ is optimized by minimizing the mse loss of the network training, which is written as

$$\mathcal{L}_{\text{train}} = \frac{1}{K}\sum_{i=1}^{K}(\hat{\mathbf{a}}_{1k} - \mathbf{a}_{1k})^2 \quad (8)$$

where $\hat{\mathbf{a}}_{1k}$ and $\mathbf{a}_{1k}$ denote the kth abundance vector in $\hat{\mathbf{A}}_1$ and $\mathbf{A}_1$, respectively.

In the second stage, we gradually reduce $\mathcal{L}_p$ to prevent candidates coupling by optimizing $\mathbf{P}^i$ and $\mathbf{Q}^j$, which is given by

$$\begin{aligned}\mathcal{L}_{\text{train}} = \frac{1}{K}\sum_{k=1}^{K}(\hat{\mathbf{a}}_{1k} - \mathbf{a}_{1k})^2 + \lambda\sum_{i,j}\sum_{\hat{i},\hat{j}}\|(\mathbf{Q}^j)^T\mathbf{Q}^{\hat{j}}\|_F^2 \\ \cdot\|(\mathbf{P}^i)^T\mathbf{P}^{\hat{i}}\|_F^2, i, j, \hat{i}, \hat{j} \in \mathcal{C}\end{aligned} \quad (9)$$

where λ is the tradeoff parameter used to control the transition of parameter sharing.

Meanwhile, the evaluation information of the channel search model on the validation set is fed back to the LSTM controller, and the controller parameters θ are updated by maximizing the expectation of the reward function $\mathcal{R}(a_c)$ using policy gradient optimization [72]. Specifically, the reward function $\mathcal{R}(a_c)$ is designed as

$$\mathcal{R}(a_c) = \frac{1}{\mathcal{L}_{\text{val}}(a_c)} \times \left[\frac{\mathcal{B}(a_c)}{B}\right]^{\eta} \quad (10)$$

where $\mathcal{L}_{\text{val}}(a_c)$ represents the mse validation loss between $\hat{\mathbf{A}}_1$ and $\mathbf{A}_1$ under different channel decisions, and η is the weight factor defined as

$$\eta = \begin{cases}\alpha, & \text{if } \mathcal{B}(a_c) \leq B \\ \beta, & \text{otherwise}\end{cases} \quad (11)$$

where α and β are coefficients of the FLOPS constraint. Note that, we alternatively minimize $\mathcal{L}_{\text{train}}$ on the training set to optimize the channel search network weight $\mathcal{W}$ and parameter sharing level $\mathcal{L}_p$ and $\mathcal{L}_{\text{val}}$ on the validation set to optimize the channel decision a_c controlled by LSTM. When the maximum numbers of steps reach or the controller parameters θ converge, the channel search network stops training and the optimal channel decision $a_c^* = (c_1, c_2, \ldots, c_l)^*$ is finally obtained based on the criterion of a minimum validation error.

C. Kernel Search Method Based on EA

Due to the insensitivity characteristics of EA to the local minima, EA has been widely used to solve complex nonconvex optimization problems [73]. Here, the genetic algorithm in EA has been adopted in this article because it is more suitable

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for solving discrete search problem, and the computational cost is relatively small. First, the search space is encoded as a sequence of numbers, such as "0" representing the convolution block with a 1×1 kernel and "3" representing the convolution block with a 7×7 kernel. Based on the encoded numbers, similar to the channel search, we randomly generate some population architectures with l layers and send them to the supernet for training.

Then, the EA is adopted to search the optimal architecture by evaluating the performance of different candidates on the validation set. The optimal kernel search is performed as

$$\upsilon^* = \arg\min_{\upsilon} \mathcal{L}_{\text{val}}(f(\upsilon, \mathbf{W}_V)) \tag{12}$$

where υ and $\mathbf{W}_V$ denote the searched kernel and the optimal supernet weight, respectively, and $\mathcal{L}_{\text{val}}$ represents the mse between $\hat{\mathbf{A}}_1$ and $\mathbf{A}_1$.

Meanwhile, several genetic operations are applied to make different offsprings from their parents. For example, the crossover operation makes candidates mix parental information to form a variety of parent-like offsprings. The mutation operation changes a random gene of candidates to make it different from their parents. Here, in order to consider the computational constraint in the search phase, a novel fitness strategy is introduced to improve the reliability of the model, which is written as follows:

$$F_i = \begin{cases} \dfrac{w_1}{\mathcal{L}_{\text{val}}(\upsilon_i)} + w_2(B - \mathcal{B}(\upsilon_i)), & \text{if } \mathcal{B}(\upsilon_i) \le B \\ \dfrac{w_1}{\mathcal{L}_{\text{val}}(\upsilon_i)}, & \text{otherwise} \end{cases} \tag{13}$$

where w_1 and w_2 are the weight coefficients to balance the validation loss and computational constraint, which are empirically set to 0.01 and 0.02, respectively. The higher fitness populations are selected as the parents, and the following crossover and mutation operations are repeated to generate enough new candidates that satisfy the given computation constraints. The specific flowchart of the proposed kernel search method based on EA is shown in Algorithm 1.

Algorithm 1 Kernel Search Based on EA for HU

Input: Validation set D_{val}, iteration number $Iter$, population size O, supernet weights $\mathbf{W}_V$, computation constraints B, mutation probability $prob$, crossover number T, mutation number M, fitness sets F,top architecture sets $Topk$, and the best architecture υ_{best}.

Outputs: The architecture with minimum loss under architecture constraints.

```
P_1 ← InitializePopulation(O, B);
T ← O/2, M ← O/2, prob ← 0.1;
while i < Iter do
    Loss_i ← Inference(W_V, P_i, D_val);
    F ← ComputeFitness(Loss_i, P_i);
    Topk ← RankFitness(F);
    P_crossover ← Crossover(Topk, T, B);
    P_mutation ← Mutation(Topk, M, prob, B);
    P_{i+1} ← P_crossover ∪ P_mutation;
end while
υ_best ←The minimum loss in Topk;
return υ_best;
```

D. Unmixing Performance Evaluation

According to the obtained network architecture, a simple AE-based unmixing network is constructed to solve the unmixing issue. Here, the mixing assumption of our approach follows the LMM, which can be formulated as

$$\begin{aligned} &\mathbf{Y} = \mathbf{S}\mathbf{A} + \mathbf{N} \\ &\text{s.t. } \mathbf{1}_S^T\mathbf{A} = \mathbf{1}_N^T, \quad \mathbf{A} \ge \mathbf{0}, \ \mathbf{S} \ge \mathbf{0} \end{aligned} \tag{14}$$

where $\mathbf{Y} \in \mathbb{R}^{L \times N}$ is the observed hyperspectral data with L bands and N pixels. $\mathbf{S} \in \mathbb{R}^{L \times S}$ is the endmember matrix with S endmember categories, and $\mathbf{A} \in \mathbb{R}^{S \times N}$ is the corresponding abundance matrix. $\mathbf{N} \in \mathbb{R}^{L \times N}$ denotes the residual matrix, containing the addictive noise and other errors. In addition, $\mathbf{A}$ satisfies the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC). $\mathbf{S}$ is also required to satisfy the endmember nonnegative constraint (ENC). $\mathbf{1}_S^T = [1, \ldots, 1] \in \mathbb{R}^{1 \times S}$ is a vector to enforce the sum of the abundance of each pixel to be 1.

The training of the AE-based unmixing network is realized by minimizing the SAD objective function between the input data $\mathbf{Y}$ and the reconstructed data $\hat{\mathbf{Y}}$, which is given by

$$\mathcal{L}_R = \frac{1}{N}\sum_{i=1}^{N} \cos^{-1}\left(\frac{\mathbf{y}_i^T\hat{\mathbf{y}}_i}{\|\mathbf{y}_i\|_2\|\hat{\mathbf{y}}_i\|_2}\right) \tag{15}$$

where $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ denote the ith vector in $\mathbf{Y}$ and $\hat{\mathbf{Y}}$, respectively.

Since the softmax function cannot produce sparse abundances, the sparsity constraint is enforced to the encoder output by introducing $L_{1/2}$ regularization. The $L_{1/2}$ regularization is expressed as

$$\mathcal{L}_{\text{reg}} = \|\mathbf{A}\|_{1/2} = \sum_{i=1}^{N}\sum_{j=1}^{S}\left|a_{ji}\right|^{1/2}. \tag{16}$$

Finally, the overall loss of the AE-based unmixing network can be formulated as

$$\mathcal{L}_{\text{AE}} = \mathcal{L}_R + \gamma\,\mathcal{L}_{\text{reg}} \tag{17}$$

where γ is the tradeoff parameter to balance the reconstruction constraint and sparsity regularization. The setting of γ on a specific dataset is fixed to ensure the fairness of comparing diverse network architectures.

IV. Experiments and Analysis

In this section, we assess the performance of the proposed method in both synthetic and real hyperspectral datasets, as shown in Fig. 3. Four classic and state-of-the-art unmixing methods are adopted to compare the proposed method, including NMF-$l_{1/2}$ [74], untied denoising AE with sparsity (uDAS) [50], deep AE unmixing (DAEU) [47], and CNN AE unmixing (CNNAEU) [52]. The GT endmembers and the corresponding abundances of the Jasper Ridge and Urban data are referenced in [75]. As for the TianGong-1 port data, due to the lack of GT, we only conduct performance analysis without quantitative comparison.

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(a) (b) (c) (d)

Fig. 3. RGB images from four hyperspectral datasets used in the experiment. (a) Synthetic data. (b) Jasper Ridge data. (c) Urban data. (d) TianGong-1 port data.

TABLE I
HYPERPARAMETERS FOR THE AUTONAS METHOD ON DIFFERENT HYPERSPECTRAL DATASETS

Phase		Synthetic	Jasper Ridge	Urban	TianGong-1
Channel Search Phase	Number of Layers	4	4	4	4
	Learning Rate of $\mathcal{W}$	1e-3	1e-3	1e-3	1e-3
	Learning Rate of θ	5e-4	2e-4	2e-4	2e-4
	Weight Decay	3e-4	1e-4	1e-4	1e-4
	Parameter Sharing Parameter λ	1e-6	1e-6	1e-6	1e-6
	FLOPs Coefficient α, β	0, -0.1	0, -0.1	0, -0.1	0, -0.1
	Initial Training Epoch	300	300	300	300
	Max Epoch	500	500	500	500
Kernel Search Phase	Number of Populations O	30	30	30	30
	Number of Top Architectures $Topk$	10	10	10	10
	Mutation Probability $prob$	0.1	0.1	0.1	0.1
	Learning Rate	8e-3	5e-3	5e-3	1e-3
	Weight Decay	5e-4	3e-4	3e-4	1e-4
	Max Epoch	200	200	200	200
Unmixing Evaluation Phase	Learning Rate	3e-3	8e-3	4e-2	8e-3
	Sparsity Regularization γ	7e-3	7e-2	7e-2	7e-2
	Max Epoch	300	500	500	500

A. Experimental Setup

1) Implementation Details: For a fair comparison, all the following experiments are executed on a computer with an Intel Core i7 and an NVIDIA GTX 1080 Ti 11-GB GPU. The number of endmembers S is determined by hyperspectral signal identification by minimum error (HySime) [76], and the initial endmembers are extracted by VCA for all comparison methods. The number of candidate endmembers is set to 2000, including 1500 used for training and 500 used for validation. Specifically, the training data are adopted to train the channel and kernel search network, while the validation data are used to evaluate the unmixing performance of different candidate architectures to obtain the optimal channel and kernel results.

The Adam optimizer [77] is adopted to update the network parameters in the training phase and the architecture constraint adopts FLOPS $\leq$ 5 G to reduce the computational complexity of the candidate architectures. In addition, to verify the effectiveness of the optimal architecture in the unmixing performance evaluation, the hyperparameters are consistent in the same dataset. The specific hyperparameter settings are shown in Table I.

2) Evaluation Metrics: In the experiment, two evaluation metrics are utilized to evaluate the precision of the estimated endmembers and the corresponding abundances, including SAD and root mse (RMSE). The SAD is defined as

$$\mathrm{SAD} = \frac{1}{S}\sum_{i=1}^{S}\cos^{-1}\left(\frac{\mathbf{s}_i^T\hat{\mathbf{s}}_i}{\|\mathbf{s}_i\|_2\|\hat{\mathbf{s}}_i\|_2}\right) \tag{18}$$

where $\mathbf{s}_i$ and $\hat{\mathbf{s}}_i$ indicate the reference endmember and the estimated endmember, respectively.

The RMSE is defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{SN}\sum_{j=1}^{N}\left\|\hat{\mathbf{a}}_j - \mathbf{a}_j\right\|_2^2} \tag{19}$$

where $\mathbf{a}_j$ and $\hat{\mathbf{a}}_j$ denote the actual abundance and the estimated abundance, respectively.

In addition, the FLOPS in the convolution layer is utilized to assess the computational complexity [78], which is given by

$$\mathrm{FLOPS} = 2H_{\mathrm{in}}W_{\mathrm{in}}\left(c_{\mathrm{in}}K_c^2 + 1\right)c_{\mathrm{out}} \tag{20}$$

where H_{in} and W_{in} denote the height and width of the input data, respectively, and K_c represents the kernel size of the convolution. Note that GFLOPS is adopted as the computation metric and GFLOPS $= 10^9\times$ FLOPS.

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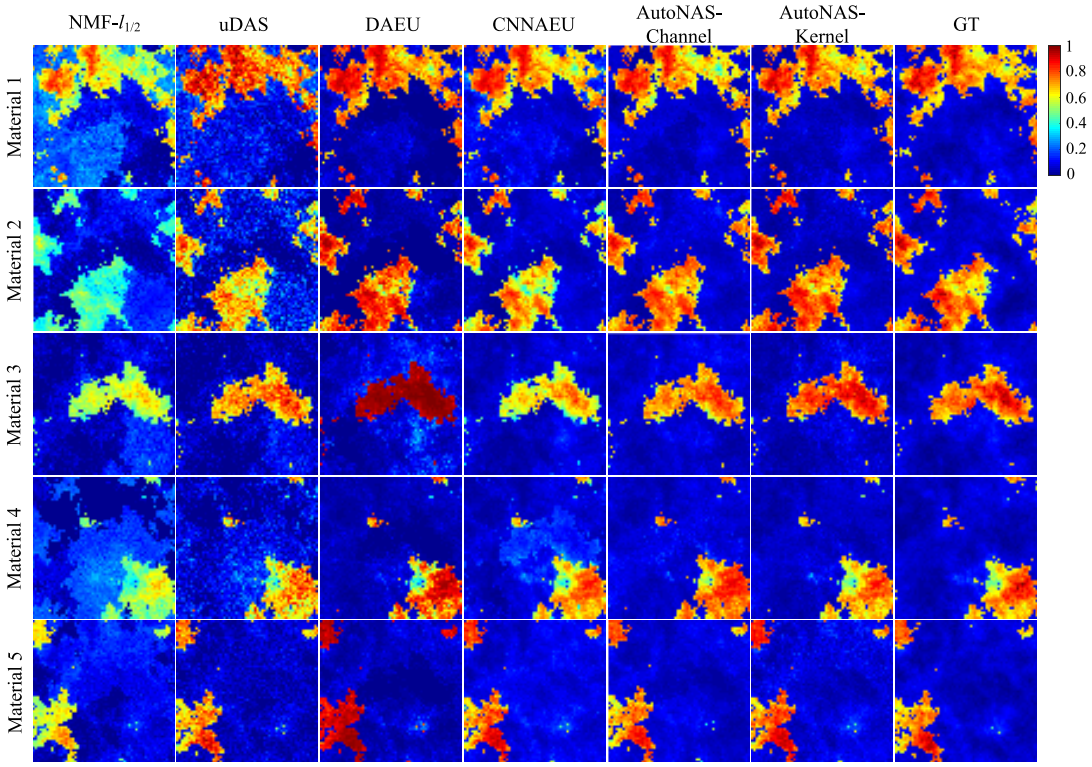


Fig. 4. Abundance maps of five materials from the synthetic data obtained by different algorithms.

TABLE II
OPTIMAL SEARCH ARCHITECTURE FOR AUTONAS

Dataset	Optimal Channel	Optimal Kernel
Synthetic	[16, 128, 96, 5]	1×1, 1×1, 1×1, 3×3
Jasper Ridge	[128, 16, 48, 4]	1×1, 1×1, 1×1, 7×7
Urban	[96, 16, 32, 4]	3×3, 5×5, 3×3, 3×3
TianGong-1 Port	[48, 32, 96, 6]	1×1, 3×3, 1×1, 1×1

TABLE III
PERFORMANCE OF THE TOP FOUR SEARCHED CHANNELS FOR AUTONAS ON THE SYNTHETIC DATA

Channel	Loss	RMSE	SAD	GFLOPs
[16, 128, 96, 5]	0.0621	0.0307	0.0113	0.04
[16, 96, 128, 5]	0.0773	0.0308	0.0193	0.04
[64, 128, 96, 5]	0.0770	0.0355	0.0129	1.72
[128, 64, 128, 5]	0.0811	0.0373	0.0166	3.44

TABLE IV
PERFORMANCE OF THE TOP FOUR SEARCHED KERNELS FOR AUTONAS ON THE SYNTHETIC DATA

Kernel	Loss	RMSE	SAD	GFLOPs
1×1, 1×1, 1×1, 1×3	0.0560	0.0299	0.0110	0.89
1×1, 5×5, 1×1, 5×5	0.0583	0.0309	0.0126	3.16
1×1, 5×5, 1×1, 7×7	0.0597	0.0320	0.0217	3.62
3×3, 1×1, 1×1, 3×3	0.0614	0.0324	0.0227	1.83

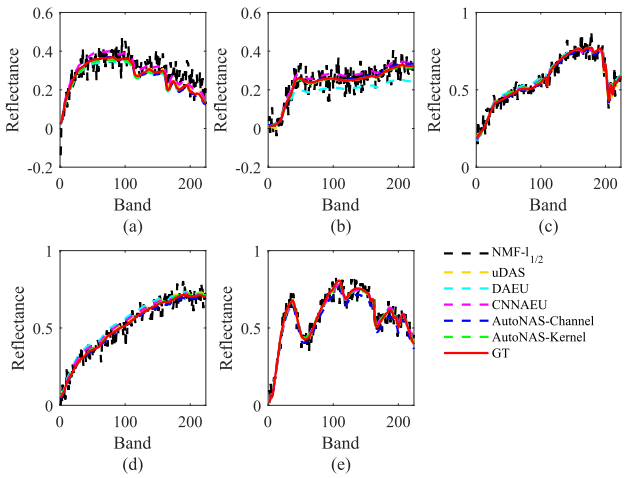


Fig. 5. Extracted endmember comparison of different unmixing algorithms in the synthetic data. (a) Material 1. (b) Material 2. (c) Material 3. (d) Material 4. (e) Material 5.

B. Experiments With Synthetic Data

Synthetic data have been widely applied to evaluate unmixing performance [70], [79]–[81]. It consists of five endmember references with 224 bands from the United States Geological Survey in the wavelength ranging from 400 to 2500 nm and the corresponding abundance maps with a size of 200×200 pixels. The abundances follow the Gaussian distribution, designed to satisfy the ANC and the ASC. Note that the synthetic data are generated based on LMM and simultaneously consider spectral variability by adding scaling factors ranging within [0.75, 1.25]. In addition, 20 dB of white Gaussian noise is added to the image to simulate the complex hyperspectral scene.

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TABLE V

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF RMSE AND SAD ON THE SYNTHETIC DATA. THE BEST RESULTS ARE SHOWN IN BOLD

Metric	NMF-$l_{1/2}$	uDAS	DAEU	CNNAEU	AutoNAS-Channel	AutoNAS-Kernel
RMSE	0.1168	0.0672	0.0676	0.0487	0.0307	**0.0299**
SAD	0.0531	0.0208	0.0259	0.0205	0.0113	**0.0110**
Running time (s)	181.40	**58.33**	77.98	129.26	6459.83	190.15

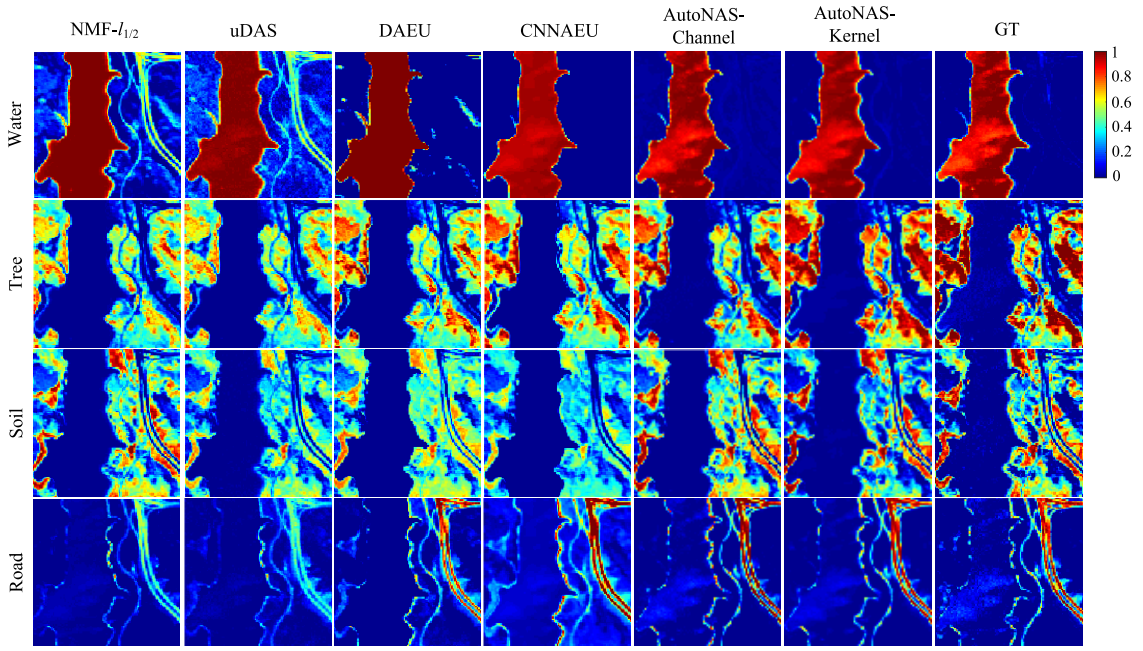


Fig. 6. Abundance maps of water, tree, soil, and road from the Jasper Ridge data obtained by different algorithms.

Table II lists the optimal search architecture for AutoNAS on the four datasets, and the specific details during searching on the synthetic data are reported in Tables III and IV. The optimal architecture is selected based on minimum validation loss. It can be seen from Tables III and IV that the top four searched architectures can basically satisfy a good unmixing trend with the decreases in validation loss, showing that the search network has been effectively trained and can achieve accurate architecture discrimination.

Table V and Figs. 4 and 5 present the quantitative performance results, the corresponding abundance maps, and the extracted endmember comparison on the synthetic data, respectively. AutoNAS-Channel and AutoNAS-Kernel in Figs. 4 and 5 represent the unmixing results of the optimal channel search when the convolutional kernel of each layer is 1 and the unmixing results of the optimal kernel search when the optimal channel is fixed, respectively. It is obvious that the DL-based unmixing methods can obtain smaller RMSE and SAD results than the traditional method, namely, NMF-$l_{1/2}$, and the extracted endmember signatures of AutoNAS basically match well with GT. The proposed AutoNAS after the channel search can achieve superior performance over the other DL-based unmixing methods, which illustrates that the DL-based methods designed by human experts are not driven by the characteristics of the hyperspectral data and the proposed channel search method based on APS can automatically search the optimal architecture. Furthermore, the kernel search results extracted by AutoNAS also outperform the channel search results, demonstrating the effectiveness of the proposed EA search method. In terms of compuational performance, AutoNAS requires a certain amount of computing resources to implement automatic search compared to

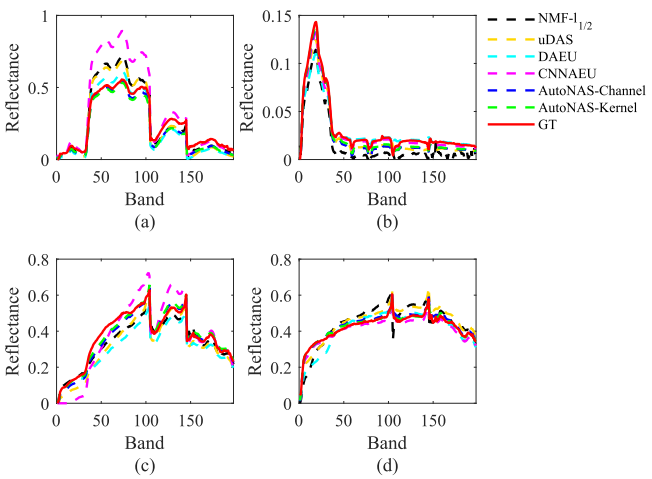


Fig. 7. Extracted endmember comparison of different unmixing algorithms in the Jasper Ridge data. (a) Tree. (b) Water. (c) Soil. (d) Road.

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TABLE VI

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF RMSE AND SAD ON THE JASPER RIDGE DATA. THE BEST RESULTS ARE SHOWN IN BOLD

Metric	NMF-$l_{1/2}$	uDAS	DAEU	CNNAEU	AutoNAS-Channel	AutoNAS-Kernel
RMSE	0.1554	0.1510	0.1156	0.1130	0.0685	**0.0447**
SAD	0.1842	0.1153	0.1270	0.0892	0.0808	**0.0716**
Running time (s)	27.76	**11.98**	13.36	25.42	1178.59	55.97

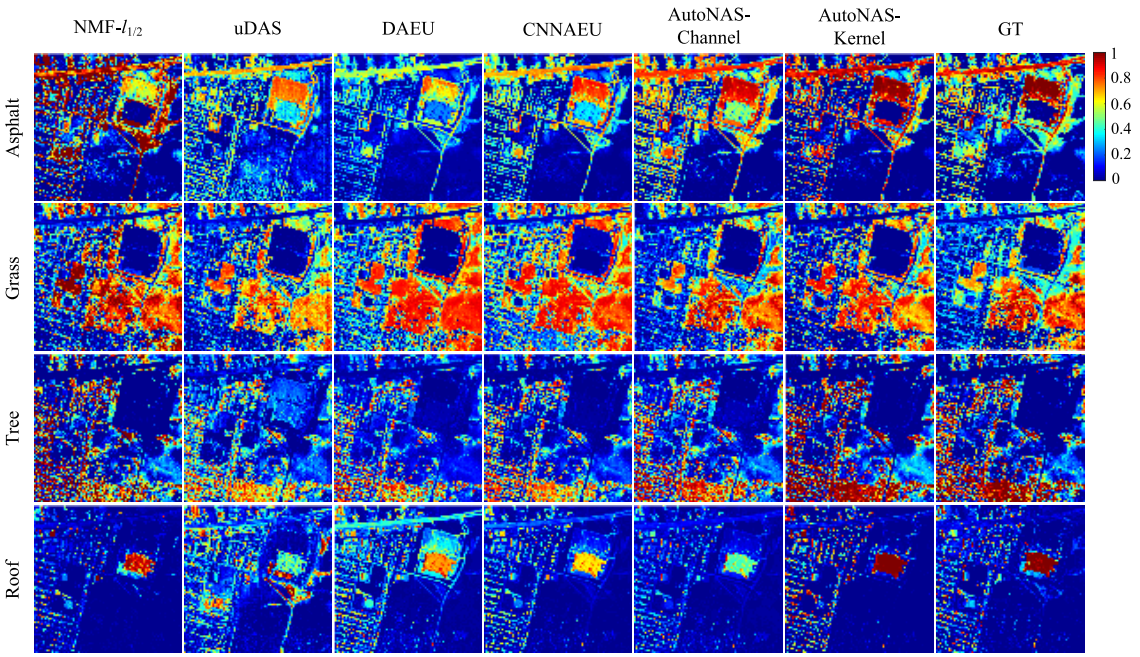


Fig. 8. Abundance maps of asphalt, grass, tree, and roof from the Urban data obtained by different algorithms.

other unmixing methods with fixed architecture because it contains training and evaluation process of different candidate architectures. On the whole, the computational cost of the automatic search technique in AutoNAS is more efficient and reasonable than that of manual design technique.

C. Experiments With Real Data

1) AVIRIS Jasper Ridge Image: The Jasper Ridge data were captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) of the Jet Propulsion Laboratory (JPL). The original image contains 512 × 614 pixels and 224 wavelength bands ranging from 380 to 2500 nm. Here, we adopt a subimage with a size of 100 × 100 pixels to evaluate the unmixing performance. After removing the channels affected by atmospheric effects and water vapor, 198 channels are used. In this studied scene, four main endmembers are investigated, that is, *#1 Water*, *#2 Tree*, *#3 Soil*, and *#4 Road*.

The quantitative performance of different methods and the estimated abundance maps on the Jasper Ridge data is shown in Table VI and Fig. 6. Overall, AutoNAS can obtain better competitive results than other handcrafted DL-based methods in terms of RMSE and SAD, demonstrating its superiority and effectiveness. For illustrative purposes, the extracted endmember comparison between AutoNAS and GT is shown in Fig. 7. Obviously, the endmember signatures extracted by AutuNAS2 are superior to those of AutoNAS,

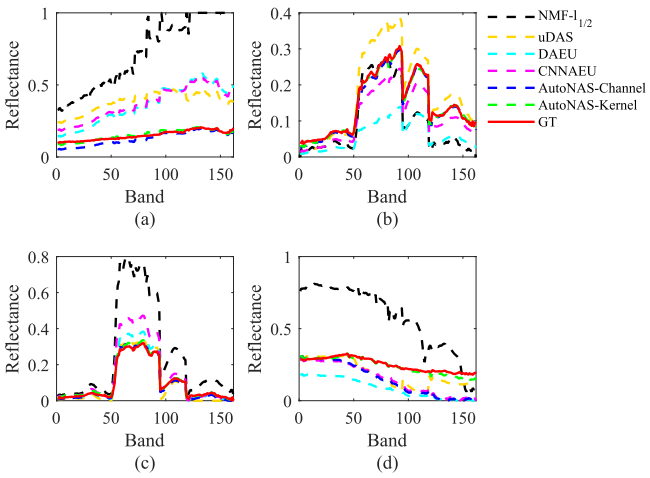


Fig. 9. Extracted endmember comparison of different unmixing algorithms in the Urban data. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof.

further illustrating that the overall AutoNAS framework can obtain more accurate and reliable unmixing results. Moreover, uDAS can achieve better computation efficiency among other comparison methods. Due to the training of the meta-weight and transformation parameters, the proposed AutoNAS in the channel search phase spends more time, but it is basically

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TABLE VII

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF RMSE AND SAD ON THE URBAN DATA. THE BEST RESULTS ARE SHOWN IN BOLD

Metric	NMF-$l_{1/2}$	uDAS	DAEU	CNNAEU	AutoNAS-Channel	AutoNAS-Kernel
RMSE	0.2368	0.2197	0.1602	0.1468	0.1071	**0.1064**
SAD	0.2544	0.2004	0.1984	0.1653	0.1835	**0.0509**
Running time (s)	382.85	**77.98**	109.94	188.63	16375.74	188.61

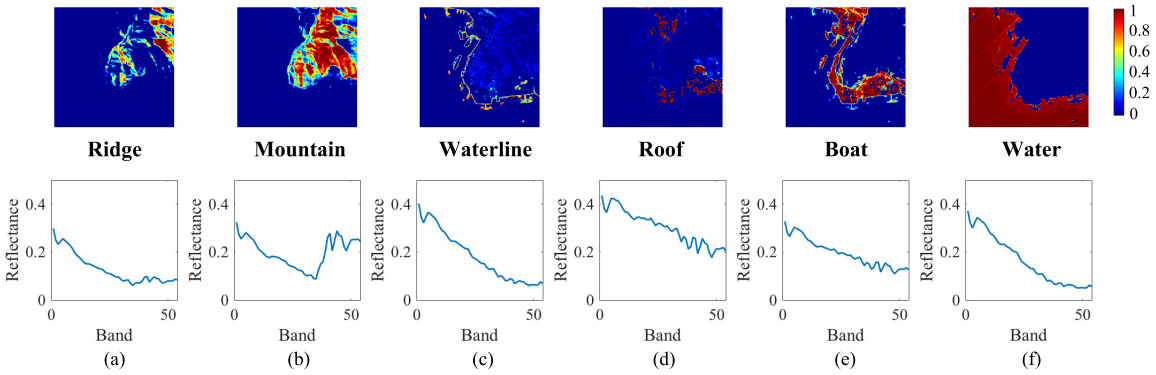


Fig. 10. Abundance maps and the corresponding endmembers extracted by the proposed AutoNAS in the TianGong-1 port data. (a) Ridge. (b) Mountain. (c) Waterline. (d) Roof. (e) Boat. (f) Water.

controlled within 20 min, indicating that the time consumption on the small dataset is relatively satisfactory.

2) HYDICE Urban Image: The Urban data were obtained by the Hyperspectral Digital Image Collection Experiment (HYDICE) sensor. The image is 307 × 307 pixels and has 210 wavelength bands, covering the spectral channels from 400 to 2500 nm. Due to the effect of water vapor and the atmosphere, only 162 bands remain. Asphalt, grass, tree, and roof are selected as the main endmembers in the experiment.

Fig. 8 visualizes the abundance maps of different methods on the Urban data, and the corresponding quantitative results are given in Table VII. As seen from Table VII, the DL-based unmixing methods can obtain more accurate endmember and abundance results than traditional methods. In addition, the corresponding abundance maps are more separable and close to the GT. Compared with the DL-based methods, the AutoNAS can fully consider the characteristics of the data to design the optimal architecture so that better unmixing results are achieved. For illustrative purposes, the extracted endmember signatures of AutoNAS are shown in Fig. 9. Obviously, the endmember signatures extracted by AutoNAS-Kernel are closer to GT compared with AutoNAS-Channel, which further proves that the proposed EA search method can obtain optimal network architecture. Furthermore, it can be seen that the AutoNAS in the kernel search phase can cost as much time as CNNAEU, demonstrating the computational effectiveness on large datasets.

3) TianGong-1 Port Image: In this section, we attempt to investigate a more complex HU scene, e.g., port, by using the proposed AutoNAS. The used HSI was collected from the TianGong-1 satellite, which was launched on September 29, 2011. The data preprocessing includes radiation correction, geometric correction, and spatial clipping, and the specific details can refer to [82]. Due to the lack of GT, the TianGong-1 data are only adopted for visual comparison rather than quantitative evaluation in order to test the performance of the proposed method in real scenes. The port image is shown in Fig. 3(d) and includes 256 × 256 pixels. After removing the channels affected by dark pixels and water vapor, there are 54 bands in the port image, covering the spectral channels from 400 to 900 nm. Six main endmembers are investigated, that is, *#1 Ridge*, *#2 Mountain*, *#3 Waterline*, *#4 Roof*, *#5 Boat*, and *#6 Water*. Fig. 10 shows the estimated abundance maps and the corresponding endmembers extracted by the proposed AutoNAS. As can be seen, AutoNAS can clearly separate multiple endmembers in the port scene, and the results indicate the potential of the proposed method in real scenarios.

V. CONCLUSION

In this article, an automatic NAS framework, called AutoNAS, has been proposed to intelligently find the most suitable AE-based architecture for HU tasks. Specifically, the proposed AutoNAS considers channel configurations and kernels simultaneously. In the channel search, APS is designed to achieve optimal performance by dynamically balancing search efficiency and architecture discrimination. Then, on the basis of the obtained optimal channels, the EA with additional computational constraints is adopted to achieve a flexible kernel search by evaluating the performance of different candidates on the validation set. Experiments with synthetic and real hyperspectral datasets demonstrate the effectiveness and superiority of the proposed method compared with several other state-of-the-art DL-based methods designed by human experts. In addition, AutoNAS can provide more fine-grained control on the computational complexity by adjusting different FLOPS constraints and help design suitable network architecture according to actual needs. In the future, we will aim to extend AutoNAS to in-orbit satellite data processing and apply to real-world problems.

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